ELECTRIC DYSFUNCTION

Flight Readiness Review

Group structure and Structures project timeline

Structures Gantt Chart

Not started
Started
Completed

		Responsibility	Due date	Pre build		Testing				Maunfacturing	
				S1 1st half	S1 2nd half	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
1	**Initiation**										
1.1	**Planning**										
	Initial design disucssion	All team members	01/11/2025								
	Open rocket designs and simulations	All team members	10/11/2025								
	Choose material for main body	All team members	01/12/2025								
	Choose material for nose cone	All team members	05/12/2025								
	Choose motor	Dan and lewis	05/12/2025								
	Decide on the rest of the materials/elements	All team members	12/12/2025								
2	**Testing and manufacturing**										
2.1	**Materials**										
	Find the materials and the parts needed	Dan	27/03/2026								
	Buy the materials and parts needed	Olivia	25/04/2026								
2.2	**First testing of each of the parts**										
	Testing the nose cone	Dan and lewis	Mar-26								
	Testing the fins	Dan	Mar-26								
	Testing the parachute	Sophie	Apr-26								
	Testing the tubing for main body	Louis and Freddie	Apr-26								
	Testing couplers	Lara and Olivia	May-26								
	Testing shock cords		May-26								
2.3	**Reviewing**										
	Review the testing	All team members	May-26								
	Change any elements if needed based on the testing	All team members	May-26								
	Redo testing if needed	All team members	May-26								
2.4	**Manufacture**										
	Cut tubing down to correct length	Dan	May-26								
	Assemble nose cone	Lewis	May-26								
	Assemble fins	Dan	May-26								
	Attach shock cords and parachute	Sophie	May-26								
	Add couplers and assemble to rocket	Lara and Olivia	May-26								

The project timeline so far shows that the rocket will be completed before launch day. Most of the assembly has been done and only has the finishing touches such as adding the logo onto the body of the rocket is left to do. The rocket is now predicted to be fully assembled and flight ready by the end of this week.

The team still consists of 14 members, and has two subteams avionics and structures. The team leads have stayed the same however the assignments for some of the team members have changed now that the project has moved onto the final assembly and decoration stages.

Name	Role	Lead	Assignment
Sam Orlin	Avionics	Avionics team lead	Hardware
Finn Woolley	Avionics		Data processing
Alistair Wye	Avionics		Software
Charlie sims	Avionics		Avionics bay & Integration
Olivia Gransden	Structures	Team lead	Assembly
Lara Burke	Structures	Vice team lead	Assembly
Louis Goodwin	Structures	Structures team lead	Assembly
Lewis Cairns	Structures		Detail and decorations
Sophie Shandley	Structures		Detail and decorations
Dan Hubbard	Structures	Vice structures lead	Assembly
Robert Seddon	Structures		Research
Freddie Kendall	Structures		Simulations
Tyler Cain	Structures		Research
James Barnet	Structures		Research

Avionics project timeline

Phase	TASK TITLE	TASK LEAD	START DATE	DUE DATE	DURATION (DAYS)	PCT OF TASK COMPLETE
1	**Procurement and Initial Code**					
1.1	All Components Procured	H&P	08/12/25	10/12/25	2	100
1.1.1	Core Sensor System Functional	H&P, S	10/12/25	10/12/25	2	100
1.2	High-Rate Logging Achieved	S	11/12/25	12/12/25	2	100
1.3	Data Pipeline Operational	S, A&V	12/12/25	26/12/25	7	100
2	**Power, Analysis & Design**					
2.05	All Components Procured	H&P	9/2/26	11/2/26	2	100
2.1	Voltage Monitoring Circuit	H&P	11/2/26	13/2/26	2	100
2.2	Critical Power Supply Tested	H&P	11/2/26	13/2/26	2	100
2.4	Full Analysis Script Complete	A&V, S	9/2/26	23/2/26	14	100
2.6	Flight Ready Schematic Finalized	H&P, M&D	16/2/26	18/2/26	2	100
	PCB acquired	H&P	15/3/26	3/4/26	19	100
3	**Manufacturing and Integration**					
3.1	Avionics Board Functional	All Team	30/03/26	02/04/26	2	100
3.2	Avionics Bay Complete	M&D	30/03/26	17/04/2007	18	100
3.2.1	Full Mission Simulation	All Team	15/04/26	17/04/26		100
3.3	GPS System Complete	M&D	15/04/26	17/04/26		80
3.2.2	Launch Readiness	All Team	TBC	TBC		100

PHASE ONE			PHASE TWO			PHASE THREE		
WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9
M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F

The avionics subsystem is near completion and on track for launch day, with all critical flight hardware successfully integrated. Since the MTR, the avionics bay has been finalized, with the PCB fully secured and mounted to the finalized 3D-printed PETG sled. The GPS system is now functional, though the ground link for the GPS coordinate system remains incomplete, and while we continue to work on it, it is considered a non-essential task for pre-flight operations and does not affect launch capability. Additionally, the LCD screen is fully operational and successfully displays maximum apogee, battery voltage, and active satellite locks.

Updated budget

6	Description	Team	Quantity	Unit cost	Shipping	Total cost	link	status
7	MicroSD Module	Avionics	1	3.4	3.9	7.3	https://thepihut.com/products/adafruit-|	Implemented
8	Microcontroller	Avionics	1	19.8	0	19.8	https://thepihut.com/products/arduino-|	Implemented
9	Barometer	Avionics	1	10.6	0	10.6	https://thepihut.com/products/adafruit-|	Implemented
10	MicroSD Card	Avionics	1	9	0	9	https://thepihut.com/products/sandisk-	Implemented
11	IMU	Avionics	1	12.5	0	12.5	https://thepihut.com/products/adafruit-|	Implemented
12	LCD	Avionics	1	8.7	0	8.7	https://thepihut.com/products/wavesha	Recieved
13	M3 Threaded Rod	Avionics	1	2.31	0	2.31	https://www.aliexpress.com/item/1005(	Implemented
14	Double SIded Foam Tape	Avionics	1	4	0	4	https://www.amazon.co.uk/dp/B0CDPL	Implemented
15	nuts n bolts	Avionics	1	8	0	8	https://www.amazon.co.uk/dp/B0F4KJ	Implemented
16	Main battery	Avionics	1	8.9	0	8.9	https://www.hobbyrc.co.uk/gnb-1500m	Implemented
17	GPS Module	Avionics	1	13.4	0	13.4	https://www.hobbyrc.co.uk/geprc-m10-	Recieved
18	ELRS Reciever	Avionics	1	11.9	0	11.9	https://www.hobbyrc.co.uk/elrs-ep2-txc	Recieved
19	22 AWG wire	Avionics	1	1.8	0	1.8	https://www.hobbyrc.co.uk/22-awg-flex	Implemented
20	GPS battery	Avionics	1	2.65	0	2.65	https://www.hobbyrc.co.uk/gnb-300ma	Recieved
21	PCB Manufacture	Avionics	5	2.25	0	2.25	https://jlcpcb.com/	Implemented
22	ABS piping	Struct...	2	14.17	9.59	37.93	https://www.plasticpipeshop.co.uk/1-12	Implemented
23	Nose cone	Struct...	1	0	0	0		Implemented
24	parachute	Struct...	1	24.98	5.24	30.22	https://modelrockets.co.uk/scratch-buil	Implemented
25	shock cord - inextensible	Struct...	3	2	5.24	11.24	https://modelrockets.co.uk/ukroc-26-cc	Implemented
26	shock cord - extendable	Struct...	3	4.3	5.24	18.14	https://modelrockets.co.uk/scratch-buil	Implemented
27	eyebolts	Struct...	1	4.49		4.49	https://www.amazon.co.uk/PATIKIL-M4	Implemented
28	spray paint	Misc	1	11.03		11.03	https://www.amazon.co.uk/Fluorescen	Recieved
29	rail guides	Struct...	1	8.1		8.1	https://modelrockets.co.uk/scratch-buil	Recieved
30	epoxy glue	Struct...	1			0	https://modelrockets.co.uk/ukroc-26-cc	Implemented
31	motor retainers	Struct...	1	3.4		3.4	https://modelrockets.co.uk/scratch-buil	Recieved
32	coupler	Struct...	1	4.79		4.79	https://modelrockets.co.uk/scratch-buil	Implemented
33	motor sleeve	Struct...	1	2.12	5.24	7.36	https://modelrockets.co.uk/scratch-buil	Recieved

The total cost of the rocket is now £259.81. All parts have now been bought and received, and the majority of parts have been implemented or are in the process of being implemented. With testing now complete, this is estimated to be the final cost of the rocket.

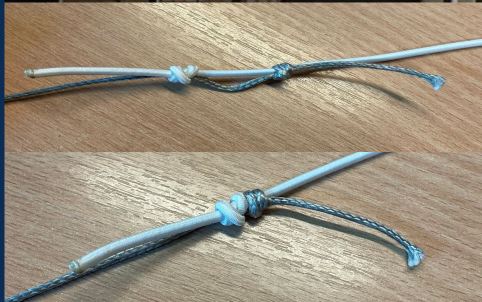
Structural strength

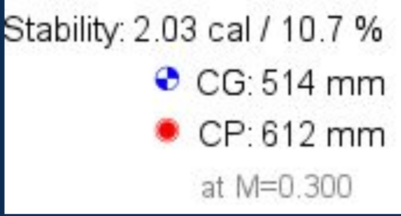
- To test the rocket’s structural integrity, a compression test was carried out on the body tube
    - At that point, the predicted G-forces on the rocket reached 10.1G - to ensure complete rocket safety the tubing was tested to 2 times the required amount, reaching 200N without any damage or flex, demonstrating the structural strength of the rocket
- Nose cone integrity has also been tested, with a drop test simulating impact speeds of 14m/s, far over the _ the rocket will experience
- Similarly, the fin assembly survivability has been tested, with a drop test demonstrating that the fins are unlikely to snap upon impact with the ground

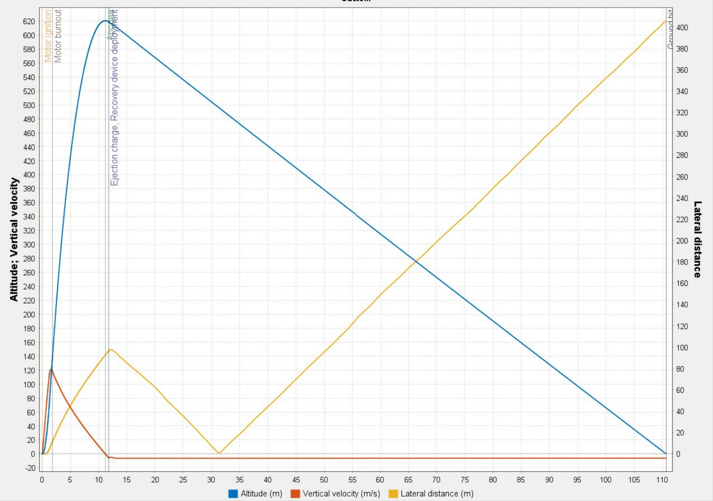
Recovery

- The recovery system has been effectively ground tested to demonstrate its effectiveness
- The parachute was drop tested with a weight equal to that of the rocket, and its descent speed was measured to ensure a low impact speed - from this we concluded that the chosen parachute is suitable for the launch
- The eye bolts have been tested by applying a sudden force to the shock cord tied to it - a slipknot was chosen to connect the shock cord due to the characteristic of it tightening when a tensile force is applied to it

- The shock cord itself was tested by fastening one end to a stationary point and pulling the other end to test its comfort under extension. From this we determined that the shock cord is highly unlikely to break during recovery system deployment. A double fisherman's knot was chosen to connect the different cord segments due to its ability of providing some shock absorption



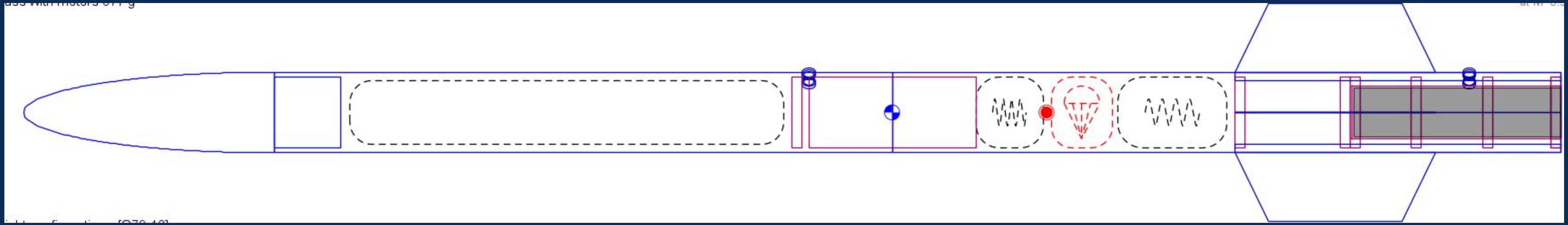




OpenRocket

- After cutting and assembling the rocket parts, the components were all weighed to get a more accurate OpenRocket Simulation to demonstrate that the design meets ARCS requirements
- This resulted in an Apogee of 621M, a stability of 2.03, an off the rod speed of 19.8m/s, and an impact speed of 8.66m/s, demonstrating the rocket is within the competition specifications

War...	Name	Configuration	Velocity off rod	Apogee	Velocity at depl...	Optimum delay	Max. velocity	Max. accelerati...	Time to apogee	Flight time	Ground hit velo...
✓	Simulation 12	[G78G/L-10]	19.8 m/s	621 m	8.66 m/s	9.4 s	121 m/s	99.6 m/s²	11.2 s	111 s	8.38 m/s



CR compliance - Structures

Specification	Requirement	Compliance
ARCS-CR-1-4-1	Teams must use safe practices when building their rockets. Construction on rockets should be done using the proper equipment and spaces such as those available in the iForge	During construction the IForge was used for cutting and sanding of the ABS piping, as well as laser cutting and construction of the fin assembly - as these were the most high risk activities it was essential they were carried out in a suitable environment. Spray painting was carried out outside with suitable protection and care. Gluing took place in a well-ventilated room.
ARCS-CR-2-1-1	Rockets must not be designed to be intentionally unstable or understable. This will be defined within this competition of having a static margin of between 1.6 and 2.2 on the launch rail	Inputting final masses into the open rocket. The stability is 2.03 off rod. which is within the required range.
ARCS-CR-2-1-3	Rockets shall sag no more than 1% of their total length when suspended from both ends	When suspended from both ends. No visible sag is apparent.
ARCS-CR-2-1-8	All rockets must be equipped with 6mm rail guides. There must be a minimum of two independent rail guides that align the rocket to the rail at launch, rail guides may not be detached during flight	The rocket has been constructed with 6mm rail guides fastened onto the side of the body tubing.

Specifications that have not been affected since CDR:

- 1-4-2
- 1-5-1
- 1-5-2
- 1-6-1
- 1-6-2
- 2-1-2
- 2-1-4
- 2-1-5
- 2-1-6
- 2-1-7
- 2-1-9
- 2-1-10
- 2-1-12
- 2-4-1

Payload Architecture

- Core Flight Computer
    - Custom 2 layer PCB driven by an ESP32 microcontroller
- High frequency logging
    - Sensor Data logged to dedicated SPI MicroSD module
- Live GNSS Tracking
    - GPS module streaming live coordinates to ground via ELRS
- Power Monitoring
    - ESP32 reads calculates battery voltage via potential divider to analog pin (fig 2)
- LCD Implementation
    - Displays Max Apogee, Battery Voltage and Satellite locks (fig 1)

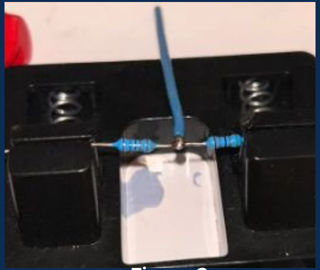
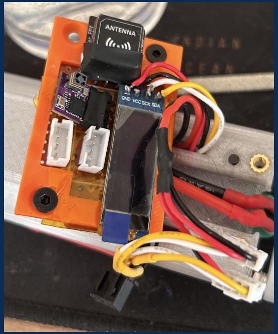


Figure 2



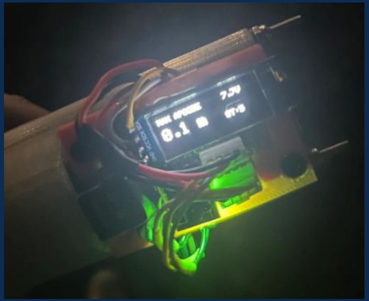
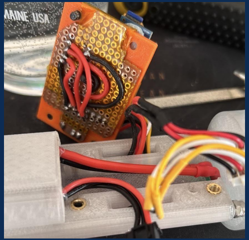
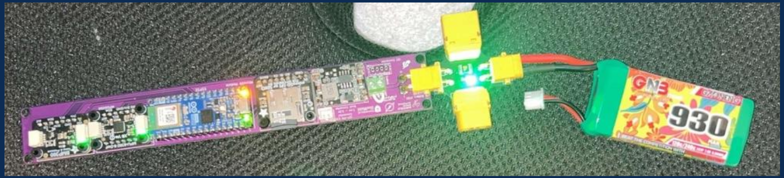


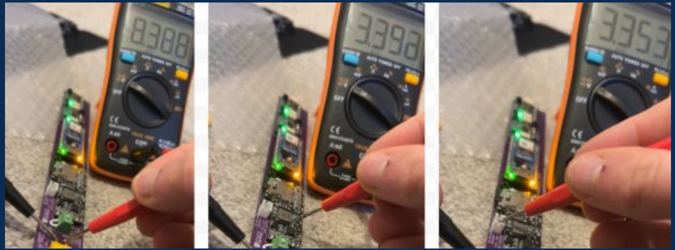
Figure 1

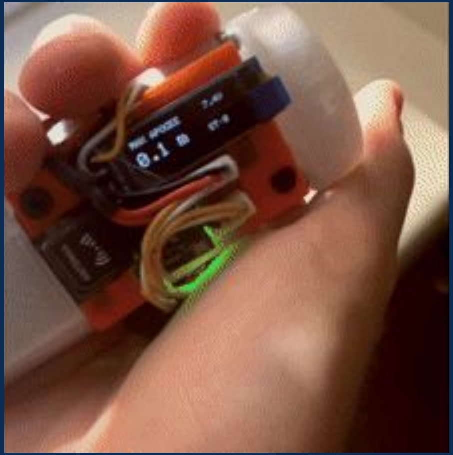


Payload Testing

- Smoke-stopper Test
  - Passed. No trip. Total isolation between live and ground verified.
- Voltage Regulation Test
  - Passed. Nominal voltages measured via multimeter. 8.4V Successfully stepped down to 3.3V.
- Battery endurance test
  - Passed. Due to lack of camera, upgraded 1500mah LiPo is highly overpowered, resulting in system logging data for multiple hours, well above the required 40 minutes.
- Telemetry and sensor verification
  - Passed. Barometer successfully logs increasing altitude when ascending stairs. Inertial measurement unit logs were passed through a 3D vector modeling tool, verifying both gyrometer and accelerometer.
- GPS System Test (NON CRITICAL)
  - Partially passed. Successfully locked onto 10+ satellites and accurately found GPS coordinates, however, link between receiver and ground station yet to be finalised.



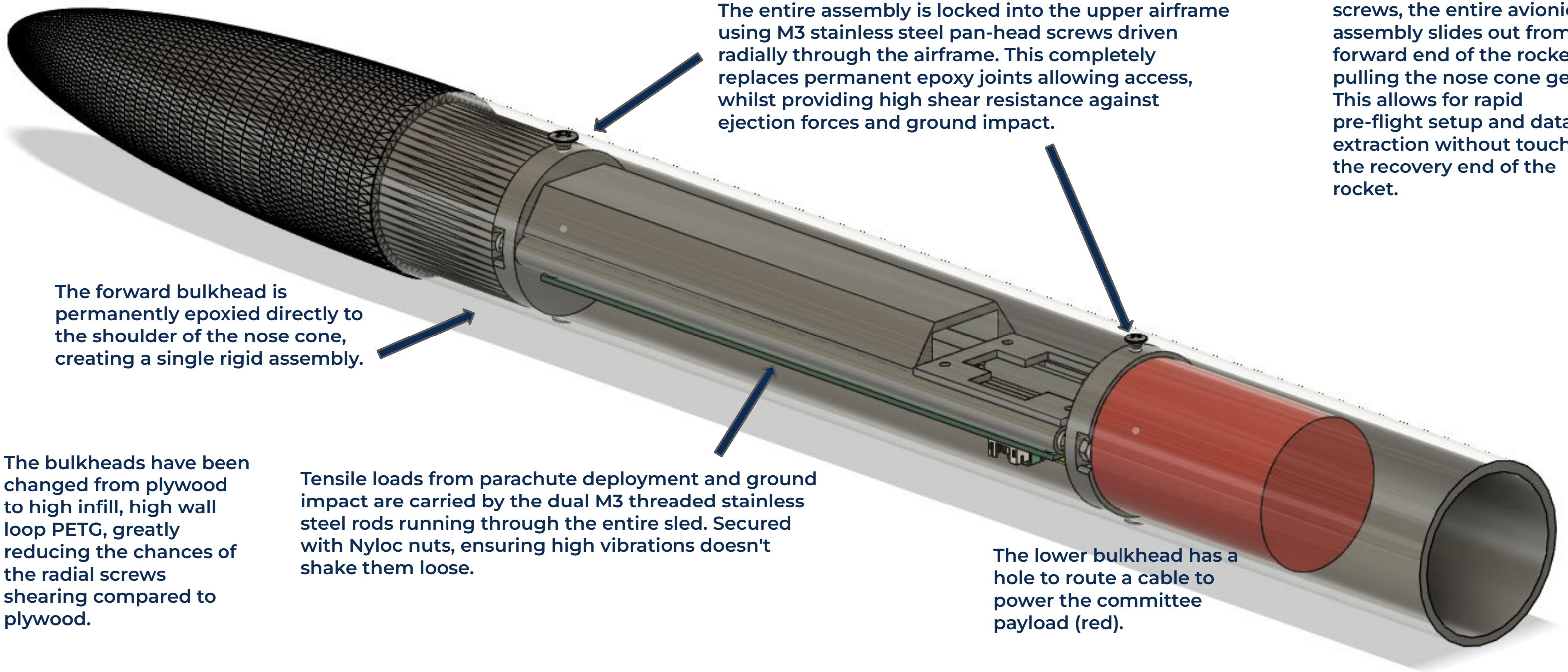




Payload Integration

The entire assembly is locked into the upper airframe using M3 stainless steel pan-head screws driven radially through the airframe. This completely replaces permanent epoxy joints allowing access, whilst providing high shear resistance against ejection forces and ground impact.

By removing the radial screws, the entire avionics assembly slides out from the forward end of the rocket by pulling the nose cone gently. This allows for rapid pre-flight setup and data extraction without touching the recovery end of the rocket.



The forward bulkhead is permanently epoxied directly to the shoulder of the nose cone, creating a single rigid assembly.

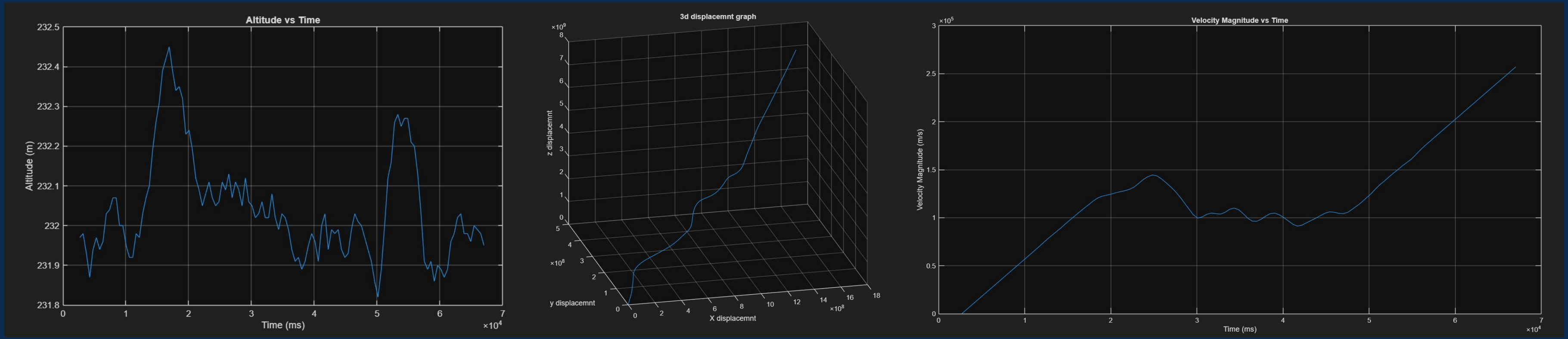
The bulkheads have been changed from plywood to high infill, high wall loop PETG, greatly reducing the chances of the radial screws shearing compared to plywood.

Tensile loads from parachute deployment and ground impact are carried by the dual M3 threaded stainless steel rods running through the entire sled. Secured with Nyloc nuts, ensuring high vibrations doesn't shake them loose.

The lower bulkhead has a hole to route a cable to power the committee payload (red).

Data display

Post recovery. The SD card will be removed from the rocket and the data will be extracted and displayed using matlab.



Graphs made from a mock set of data from testing.

Avionics CR Compliance

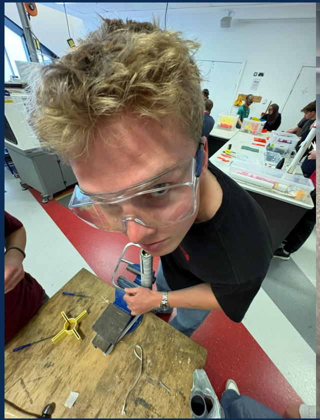
Specification	Requirement	Compliance
ARCS-CR-2-2-1	All teams must ensure the place is well ventilated before soldering any component. Additionally, all teams must ensure that all solder used is lead-free.	All soldering performed in ventilated spaces using lead-free solder.
ARCS-CR-2-2-2	Before connecting a battery to your components, verify that no short circuits are present and everything is routed correctly. We suggest using a smoke-stopper when first testing avionics. Ensure that the battery is not routed to the USB port of your device.	The battery is never routed to the USB port; electrical continuity tests are in place. A smoke stopper will be used.
ARCS-CR-2-2-3	Batteries above 7.4V are not allowed.	We will use a 2S 7.4V LiPo battery.
ARCS-CR-2-2-4	It is recommended to use multi-core wires for signal connections. Connectors such as JST and XT are recommended as well.	Use of multi-core wiring for all flexible connections to prevent fatigue. Using XT30 connector for our battery.
ARCS-CR-2-2-5	All teams must mechanically mount (i.e. with screws & bolts) their PCB; hot glue or epoxy can't be used to stick your avionics to the avionics bay. PCB mounts must be designed to resist the shock forces experienced by the rocket during its flight.	Our PCB mounts to the 3D printed sled with 11 screws into heat set threaded inserts. Our avionics bay uses rubber o rings between bulkhead and sled to reduce the effects of vibration. M3 stainless steel rods run axially, through the avionics bay, fronting the tensile loads, and the bay is screwed in radially, providing strong shearing strength.

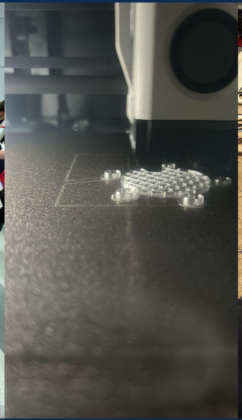
Decoration

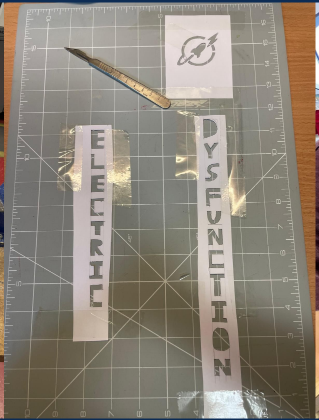
- The colours chosen to paint the rocket have been altered in favour of maximising visibility to aid in rocket recovery. For this reason the body tube was sprayed with a colour gradient of red, neon pink and neon orange, with the nose cone and fins being sprayed with blue
- The team name and logo will be painted along the body in blue to match the nose cone and fins.
- To ensure better adhesion to surfaces, a layer of primer was applied - this also increased the brightness of the paint sprayed on top of it, further helping with recovery

Last steps of assembly

- Drill holes for barometer(this will done on 02/06)
- Paint lettering using stencil
- Screw in rail guides(will be done after painting the lettering)







TEAM PHOTOS